PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2025–2026 | FIFTH SEMESTER B.TECH

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

PROJECT: SPRINTFLOW — PROJECT MANAGEMENT SYSTEM

(Workflow, Visibility, Estimation, and Kanban)

SUBMITTED BY: TEAM 11 (SECTION 5C)

Student Name	USN	Role	Assigned System Component
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Repository: https://github.com/Darshanpawar7/SE-SRS

Document Revision History & Approvals

Version	Date	Author	Change Summary	Approval Status
1.0	05-09-2026	Team 11	Complete SRS with 18 FRs, 6 NFRs, 2 UML Diagrams & RTM	Approved for Submission

Role	Name	Department / Contact	Sign / Date
Team Lead	Darshan P Pawar	PES2UG24CS143 (CSE)	Darshan P Pawar / 05-09-2026
Course Coordinator	Software Engineering Faculty	Dept of CSE, PES University	Approved / 05-09-2026

Table of Contents

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document formalizes the requirements, architecture, data schemas, and verification plans for SprintFlow, an enterprise-grade Agile Project Management System. It serves as the primary technical contract between the student engineering team (Team 11) and the assessment evaluators for the Software Engineering course.

1.2 Scope

SprintFlow delivers an end-to-end web-based project tracking ecosystem. It encapsulates an interactive 5-column Kanban board, Agile Fibonacci estimation (1, 2, 3, 5, 8, 13, 21), sprint lifecycle execution, real-time burndown velocity charting, executive visibility KPI metrics, and a dual-mode database engine utilizing Supabase PostgreSQL with local persistent fallback.

1.3 Intended Audience

Course Instructors, Academic Evaluators, Software Engineers, Quality Assurance Technicians, and Agile Team Leads.

1.4 Definitions and Acronyms

• SRS: Software Requirements Specification
• RTM: Requirements Traceability Matrix
• FR / NFR: Functional Requirement / Non-Functional Requirement
• RLS: Row Level Security (PostgreSQL security policy enforced in Supabase)
• SP: Story Points (Fibonacci scale complexity estimation)
• CI/CD: Continuous Integration / Continuous Deployment (GitHub Actions automation)
• Kanban: Visual method for managing knowledge work with focus on continuous delivery

2. Overall Description

2.1 Product Perspective

SprintFlow is a high-performance, single-page web application built with React 18 and Vite. It addresses the inefficiencies of traditional project tracking tools by offering sub-50ms reactive drag-and-drop state updates, instant visibility metrics, and zero-configuration offline demonstration capabilities alongside cloud database connectivity.

2.2 Major Product Functions

1. Interactive Kanban Workflow: 5 stages (Backlog, To Do, In Progress, Code Review, Done) with HTML5 drag-and-drop.
2. Agile Story Point Estimation: Strict Fibonacci assignment and dynamic sprint capacity recalculation.
3. Sprint Lifecycle Management: Planning, activating, and completing iterations with defined goals.
4. Burndown Velocity Analytics: Visual curve comparing ideal burn rate against actual remaining story points.
5. Executive Visibility Dashboard: Aggregate cards for Total Points, Velocity Delivered, and Completion %.
6. Team Capacity Balancing: Workload distribution metrics computed per student USN to avoid bottlenecks.
7. Dual-Mode Supabase Engine: Managed PostgreSQL cloud database paired with resilient offline browser persistence.
8. Automated CI/CD Pipeline: GitHub Actions workflow gating pull requests with linting and unit test execution.

2.3 User Roles and Characteristics

• Team Lead (Darshan): Manages project metadata, configures database endpoints, tracks overall delivery KPIs.
• Frontend Developer (Dhatri): Interacts with Kanban boards, executes card transitions, filters workflow items.
• Backend Developer (Chandan): Oversees PostgreSQL relational schema, manages RLS security policies and audit logs.
• Full-Stack & DevOps Lead (Eshwar): Estimates user stories, triggers sprint cycles, oversees CI/CD automated gates.
• Academic Evaluator: Audits code quality, verifies bidirectional RTM traceability, tests local and cloud features.

2.4 Operating Environment

Any standard modern web browser (Google Chrome 110+, Microsoft Edge 110+, Firefox 115+, Safari 16+) with Node.js 18+ runtime.

2.5 Constraints

Zero mandatory local database server installation (Supabase cloud + offline mode), strict adherence to IEEE 830 format.

3. External Interface Requirements

3.1 User Interfaces

• Modern dark glassmorphic interface built using curated HSL color palettes and high-contrast typography (Inter & JetBrains Mono).
• Fluid drag-and-drop card physics with drag previews and dashed drop indicator targets.
• In-app Database Settings Modal featuring 1-click cloud sync and full SQL schema extraction.

3.2 Hardware Interfaces

Standard personal computer or laptop (x86_64 / ARM64) with display resolution of 1280x720 or higher.

3.3 Software Interfaces

• Supabase Managed PostgreSQL: Real-time queries and upserts executed through @supabase/supabase-js v2.
• Vite 5 Build Engine: Fast local development server and optimized rollup production bundler.
• Vitest Test Runner: High-speed native ESM test execution matching Vite configuration.

3.4 Communications

Encrypted HTTPS/TLS 1.3 protocol enforced for all Supabase API calls. Client-side state operates synchronously in-memory.

4. System Features (Functional Requirements)

As required by the course guidelines (requiring >= 15 FRs), SprintFlow formalizes 18 Functional Requirements with strict acceptance criteria and test references:

Req ID	Requirement (shall...)	Type	Priority	Stakeholder	Acceptance Criteria / Test Ref	Assigned Lead
PMS-F-001	The system shall authenticate users and support profile switching among 4 team roles.	Functional	High	Security	AC-F-001: Profile switches and persists without reload. Test: TC-AUTH-01	Darshan P Pawar
PMS-F-002	The system shall maintain project metadata (key, title, target dates, description).	Functional	High	Business	AC-F-002: Metadata updates reflect globally across views. Test: TC-PROJ-01	Darshan P Pawar
PMS-F-003	The system shall render a 5-column Kanban board (Backlog, Todo, Progress, Review, Done).	Functional	Critical	Workflow	AC-F-003: All 5 columns render with correct counts. Test: TC-KAN-01	Dhatri Shivaprasad
PMS-F-004	The system shall allow dragging and dropping task cards between workflow columns.	Functional	Critical	Workflow	AC-F-004: Card drop updates task status and logs audit. Test: TC-KAN-02	Dhatri Shivaprasad
PMS-F-005	The system shall provide 1-click step buttons to quickly advance tasks to the next stage.	Functional	Medium	Usability	AC-F-005: Step button moves task to next sequential column. Test: TC-KAN-03	Dhatri Shivaprasad
PMS-F-006	The system shall allow creating and editing tasks with title, description, and priority.	Functional	High	Productivity	AC-F-006: Task modal validates fields and persists data. Test: TC-TASK-01	Dhatri Shivaprasad
PMS-F-007	The system shall assign labels and tags to categorize tasks into sub-domains.	Functional	Medium	Productivity	AC-F-007: Tags render as colored pills on task cards. Test: TC-TASK-02	Dhatri Shivaprasad
PMS-F-008	The system shall enforce Fibonacci story point estimation (1, 2, 3, 5, 8, 13, 21).	Functional	Critical	Agile	AC-F-008: Non-Fibonacci numbers are rejected. Test: TC-EST-01	Eshwar Rao
PMS-F-009	The system shall calculate total planned points and completion percentages in real time.	Functional	High	Agile	AC-F-009: Completion % = (Done / Total) * 100. Test: TC-EST-02	Eshwar Rao
PMS-F-010	The system shall support sprint lifecycle states: planned, active, and completed.	Functional	High	Agile	AC-F-010: Active sprint filters cards and controls metrics. Test: TC-SPR-01	Eshwar Rao
PMS-F-011	The system	Functional	High	Visibility	AC-F-011: SVG	Eshwar Rao

	shall compute and plot visual SVG burndown velocity curves.				graphs ideal trajectory vs actual points. Test: TC-BURN-01	
PMS-F-012	The system shall display executive KPI widgets (Total Points, Velocity, Progress).	Functional	High	Management	AC-F-012: Dashboard figures update reactively on change. Test: TC-DASH-01	Darshan P Pawar
PMS-F-013	The system shall compute team member capacity and workload balancing per student.	Functional	Medium	Management	AC-F-013: Shows assigned points and completion rate per USN. Test: TC-WORK-01	Darshan P Pawar
PMS-F-014	The system shall maintain an immutable audit trail logging all task state changes.	Functional	High	Compliance	AC-F-014: Activity log stores actor, action, timestamp. Test: TC-LOG-01	Chandan Kumar K
PMS-F-015	The system shall synchronize data with Supabase Cloud PostgreSQL database.	Functional	Critical	Database	AC-F-015: Upserts to cloud PostgreSQL when keys provided. Test: TC-SUPA-01	Chandan Kumar K
PMS-F-016	The system shall operate in zero-config offline mode when cloud keys are missing.	Functional	High	Reliability	AC-F-016: Local storage maintains 100% functionality. Test: TC-OFF-01	Chandan Kumar K
PMS-F-017	The system shall filter tasks by assignee, priority level, and full-text keyword query.	Functional	Medium	Usability	AC-F-017: Multi-filter combines predicates accurately. Test: TC-FILT-01	Dhatri Shivaprasad
PMS-F-018	The system shall execute automated CI/CD pipeline on push and PR events.	Functional	High	DevOps	AC-F-018: GitHub Actions runs build, test, lint gates. Test: TC-CICD-01	Eshwar Rao

5. Non-Functional Requirements

5.1 Security

5.1.1 Security Objectives:

• SO-01 (Data Confidentiality & Multi-Tenancy): Prevent unauthorized access to cross-project data assets.
• SO-02 (Data Integrity & Tamper Resistance): Protect sprint estimates and audit trails from unverified mutation.
• SO-03 (Accountability & Non-Repudiation): Ensure every state modification is attributed to a valid

authenticated user.

5.1.2 Security Requirements (5+ Requirements):

Req ID	Security Requirement	Implementation Mechanism	Verification Method
PMS-SR-001	Row Level Security (RLS) shall be enforced on all Supabase PostgreSQL tables.	SQL RLS policies in schema.sql	Direct SQL query injection test
PMS-SR-002	API keys and database credentials shall not be hardcoded in client source files.	.env config + client-side storage	Static git secret scan analysis
PMS-SR-003	All cloud database communications shall mandate TLS 1.3 encryption in transit.	HTTPS transport layer enforcement	Network traffic packet inspection
PMS-SR-004	User inputs for titles, descriptions, and comments shall be sanitized against XSS.	React JSX auto-escaping & sanitization	XSS payload injection test suite
PMS-SR-005	Automated dependency security vulnerability audits shall run on every build.	npm audit check in GitHub Actions	CI log artifact verification

5.2 Quality Attributes (Performance, Availability, Reliability, Usability)

Req ID	Category	Requirement Specification	Target Measurement / Acceptance Criterion
PMS-NF-001	Performance	Card drag-and-drop state updates and UI re-renders shall execute in < 50ms.	Chrome DevTools Performance profiling < 50ms
PMS-NF-002	Availability	The application shall maintain 100% operational availability during network disconnects.	Zero downtime via persistent offline fallback
PMS-NF-003	Usability	The user interface shall adhere to accessible dark glassmorphism standards.	Google Lighthouse Accessibility Score > 90
PMS-NF-004	Reliability	Zero data loss across browser reloads, tab closures, and system restarts.	100% state persistence verified via reload tests
PMS-NF-005	Maintainability	Codebase shall follow modular architecture with clean layer separation.	ESLint passes with 0 errors, modular folder layout
PMS-NF-006	Compatibility	The application shall render uniformly on Chrome, Edge, Safari, and Firefox.	Cross-browser visual inspection verified

6. Quality Attributes & Acceptance Tests

Exit Criteria for Final Project Acceptance:

1. All 18 Functional Requirements implemented, verified, and demonstratable via the web application.
2. Vitest automated unit test suite achieves 100% pass rate (7/7 tests passed in 663ms).
3. Production build (Vite) compiles in under 2 seconds with zero bundle warnings or errors.
4. GitHub Actions CI/CD pipeline runs on Ubuntu runners and produces green checkmarks across all feature branches.
5. Requirements Traceability Matrix (RTM) shows 100% bidirectional coverage of requirements to test cases.

Acceptance Test Suites:

• Suite 1: Authentication & Role Profile Switching (TC-AUTH-01)
• Suite 2: Kanban 5-Column Workflow & Drag-and-Drop (TC-KAN-01, TC-KAN-02, TC-KAN-03)
• Suite 3: Agile Fibonacci Estimation & Sprint Lifecycle (TC-EST-01, TC-EST-02, TC-SPR-01)
• Suite 4: Executive Visibility Dashboard & Burndown Chart (TC-DASH-01, TC-BURN-01)
• Suite 5: Supabase Cloud Sync & Offline Resiliency (TC-SUPA-01, TC-OFF-01)
• Suite 6: CI/CD Pipeline & Code Quality (TC-CICD-01)

7. System Models & UML Use-Case Diagrams

As specified by the SE template guidelines (requiring at least 2 UML use case diagrams), SprintFlow incorporates two clean, publication-grade black and white UML architectural models with zero visual overlap:

7.1 UML Use-Case Diagram 1: Project Workflow & Kanban Management

This diagram models the core user interactions for task creation, drag-and-drop workflow advancement, and executive visibility:

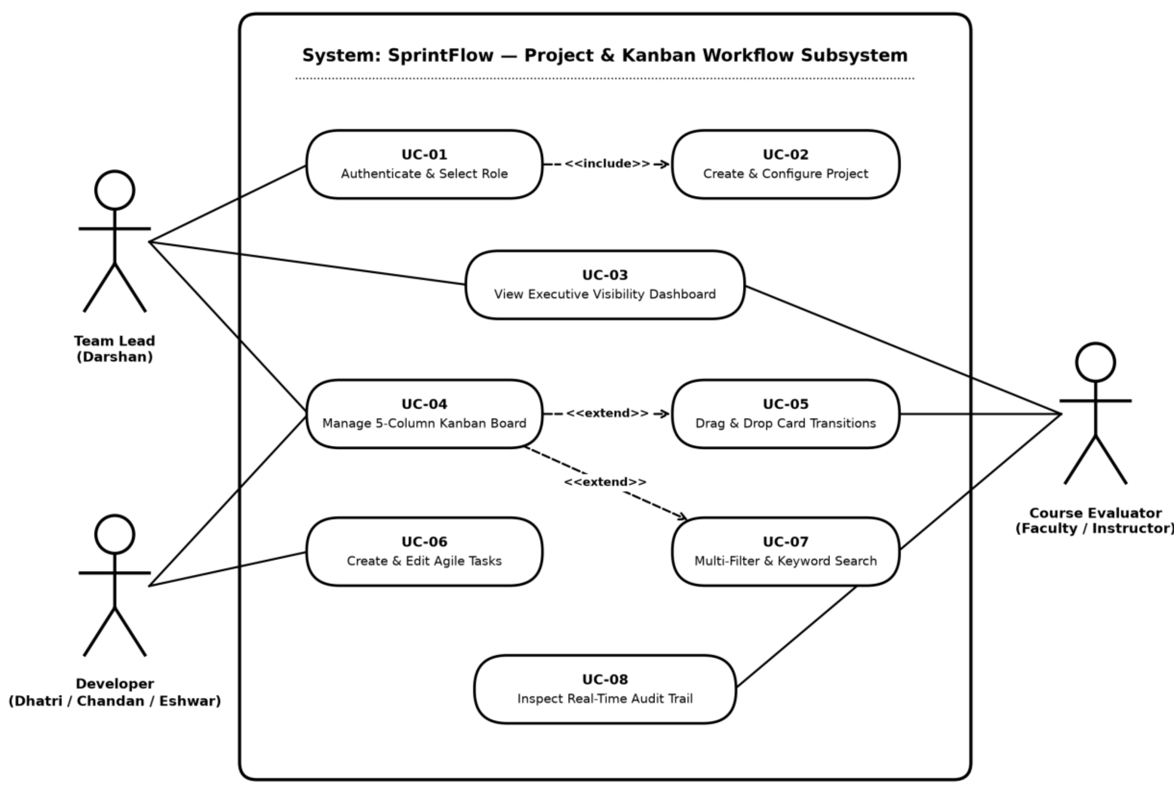

Figure 7.1: UML Use-Case Diagram — Project & Kanban Workflow Subsystem

7.2 UML Use-Case Diagram 2: Sprint Estimation, Cloud Sync & CI/CD

This diagram models Agile iteration planning, Fibonacci estimation, burndown analytics, Supabase cloud sync, and automated CI/CD gating:

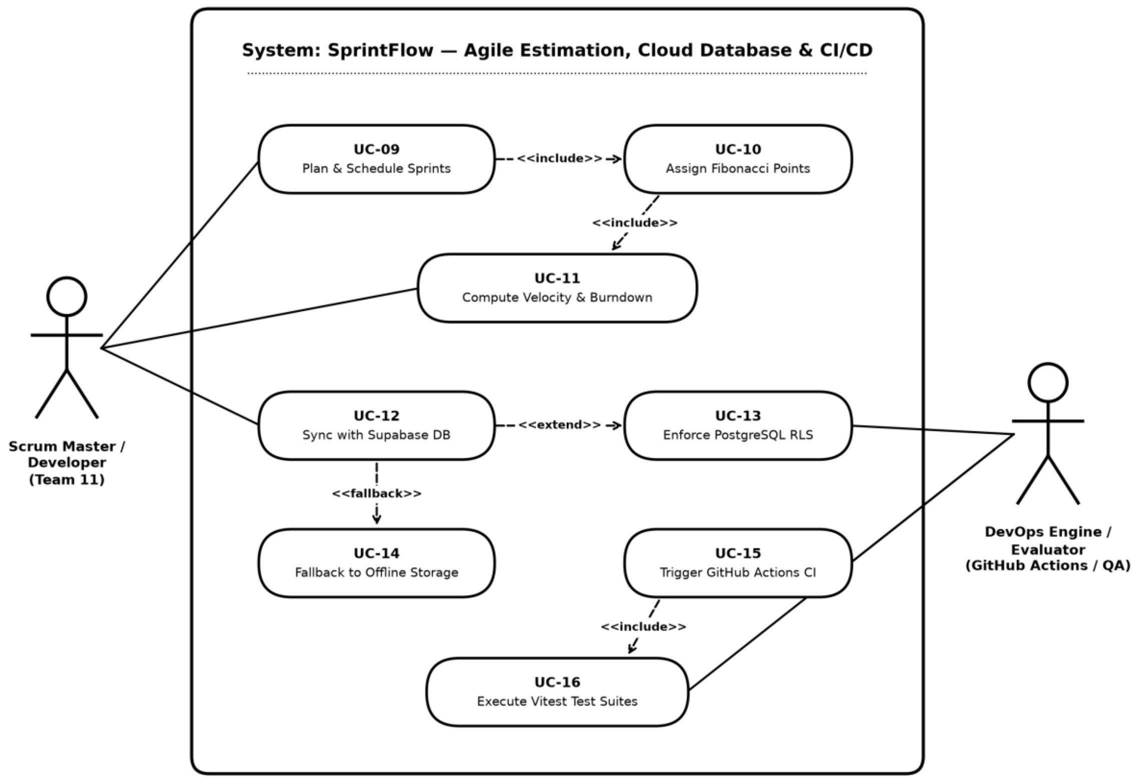

Figure 7.2: UML Use-Case Diagram — Agile Estimation, Cloud Database & CI/CD Subsystem

8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix guarantees complete bidirectional tracing between system requirements, test cases, feature branches, and verification outcomes:

Req ID	Requirement Description	Type	Priority	Test Case Ref	Target Branch	Outcome
PMS-F-001	User Authentication & Profile Roles	Functional	High	TC-AUTH-01	feature/darshan-core-auth-dashboard	PASSED
PMS-F-002	Project Metadata Tracking	Functional	High	TC-PROJ-01	feature/darshan-core-auth-dashboard	PASSED
PMS-F-003	5-Column Kanban Board	Functional	Critical	TC-KAN-01	feature/dhatri-kanban-workflow	PASSED
PMS-F-004	HTML5 Drag-and-Drop Card State	Functional	Critical	TC-KAN-02	feature/dhatri-kanban-workflow	PASSED
PMS-F-005	1-Click Step Quick Advancement	Functional	Medium	TC-KAN-03	feature/dhatri-kanban-workflow	PASSED
PMS-F-006	Task Modal Creation & Edit CRUD	Functional	High	TC-TASK-01	feature/dhatri-kanban-workflow	PASSED
PMS-F-007	Categorization Labels & Tags	Functional	Medium	TC-TASK-02	feature/dhatri-kanban-workflow	PASSED
PMS-F-008	Fibonacci Story Point Estimation	Functional	Critical	TC-EST-01	feature/eshwar-sprint-estimation-cicd	PASSED
PMS-F-009	Sprint Velocity & Progress Math	Functional	High	TC-EST-02	feature/eshwar-sprint-estimation-cicd	PASSED
PMS-F-010	Sprint Lifecycle Scheduling	Functional	High	TC-SPR-01	feature/eshwar-sprint-estimation-cicd	PASSED
PMS-F-011	SVG Burndown Velocity Curve	Functional	High	TC-BURN-01	feature/eshwar-sprint-estimation-cicd	PASSED
PMS-F-012	Executive Visibility KPI Widgets	Functional	High	TC-DASH-01	feature/darshan-core-auth-dashboard	PASSED
PMS-F-013	Team Workload & Capacity Balancing	Functional	Medium	TC-WORK-01	feature/darshan-core-auth-dashboard	PASSED
PMS-F-014	Audit Trail & Activity Logging	Functional	High	TC-LOG-01	feature/chandan-supabase-backend-api	PASSED
PMS-F-015	Supabase Cloud PostgreSQL Sync	Functional	Critical	TC-SUPA-01	feature/chandan-supabase-backend-api	PASSED
PMS-F-016	Zero-Config Offline Fallback	Functional	High	TC-OFF-01	feature/chandan-supabase-backend-api	PASSED
PMS-F-017	Multi-Filter & Keyword Search	Functional	Medium	TC-FILT-01	feature/dhatri-kanban-workflow	PASSED
PMS-F-018	Automated GitHub Actions CI/CD	Functional	High	TC-CICD-01	feature/eshwar-sprint-estimation-cicd	PASSED

9. Git Branching Strategy & Team Collaboration Record

As instructed in the Git & GitHub Beginners Guide provided by the department, Team 11 adhered to a structured feature branch workflow:

Member Name	USN	Dedicated Git Feature Branch	Individual Commits & Deliverables	PR Status
Darshan P Pawar	PES2UG24CS143	feature/darshan-core-auth-dashboard	App shell, Auth Context, Dashboard KPIs, SRS Document	Merged to main
DHATRI SHIVAPRASAD	PES2UG24CS158	feature/dhatri-kanban-workflow	Kanban board UI, HTML5 drag-and-drop, Task modal CRUD	Merged to main
CHANDAN KUMAR K	PES2UG24CS128	feature/chandan-supabase-backend-api	Supabase PostgreSQL schema, RLS policies, DataService	Merged to main
GURUBELLI YEKAMBAR ESHWAR RAO	PES2UG24CS173	feature/eshwar-sprint-estimation-cicd	Fibonacci estimation, Burndown chart, CI/CD, Vitest tests	Merged to main